

Experiential Learning

Report submitted to the

VIT Bhopal University

B.tech

in

Computer Science and Engineering

(Artificial Intelligence and Machine Learning)

Submitted by

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Declaration

I, ***Shrish***, bearing the Registration Number ***23BA111284*** hereby declare that this report of “*Experiential Learning*” represents my original work carried out as a undergraduate student at VIT Bhopal University. To the best of my knowledge, it contains no material previously published or written by another person, nor any material presented for the award of any other degree of VIT Bhopal University or any other institution. Any contribution made to this report by others, with whom I have worked at VIT Bhopal University or elsewhere, is explicitly acknowledged in the report.

16 November, 2025

Shrish

VIT Bhopal University

Acknowledgment

I would like to express my sincere gratitude to **VIT Bhopal University** for providing me with the opportunity to participate in the **Experiential Learning Program for the academic year 2025–2026**. This program has been instrumental in enhancing my practical understanding, problem-solving ability, and real-world application of academic concepts.

I extend my heartfelt thanks to the **Hon'ble Vice Chancellor**, the **Dean**, and the **Faculty Coordinators** for their continuous guidance, motivation, and support throughout the learning journey. Their commitment to fostering innovation and experiential learning has been truly inspiring.

I am also grateful to my **mentors, instructors, and peers** whose collaboration and insights enriched my learning experience. The hands-on activities, interactive sessions, and industry-oriented approach of this program have significantly contributed to my personal and professional growth.

Finally, I thank everyone involved in organizing and managing this program for making it a valuable and transformative learning experience.

Shrish

Summary of Your Experiential Learning

As part of the **VIT Bhopal Experiential Learning Program**, I had the opportunity to engage in a series of industrial visits, expert interactions, and practical learning sessions across **Chennai** and **Bangalore**. These experiences significantly enhanced my understanding of emerging technologies, industrial operations, and real-world applications.

Chennai Visit

During the Chennai segment of the program, I visited **VIT Chennai**, where I attended an **expert session** focusing on industry expectations, technology trends, and employability skills.

On the **second day**, I attended a learning session at **TVS**, gaining exposure to automotive manufacturing processes, quality control, and industrial workflow.

On the **third day**, I participated in an **IoT (Internet of Things)** at **VIT Chennai** session that provided insights into sensor technology, embedded systems, and real-time applications.

On the **fourth day**, visited to **BSNL** offered insights into telecommunication networks, switching systems, broadband infrastructure, and communication technologies.

On the **fifth day**, visited to **NIOT**, where I learnt engineering applications in oceanography, deep-sea exploration, and marine resource utilization.

Bangalore Visit

The Bangalore segment began with **two days of training at Cranes Varsity**. The sessions covered high-performance computing, Machine Learning fundamentals.

On the **third day**, I visited the **NeoSky**, where I Gained knowledge of drone components, materials used, and the engineering behind UAV construction.

On the **fourth day**, I visited a **science and technology museum in Bangalore**, which provided hands-on exposure to mechanical models, electronic systems, optical devices, and historical technological innovations.

On the **fifth day**, we had an **interactive session with industry experts**, gaining a deeper understanding of current industrial trends, career paths, and future technological advancements.

Overall, this experiential learning journey enriched my technical knowledge, improved my industry awareness, and strengthened my practical learning skills.

Contents

	Description	Page No.
	Declaration of originality.....	
	Acknowledgment.....	
	Summary	
	Contents.....	
Chennai	Day – 1 report	
	1.1.1. V.I.T Chennai.....	
	1.1.2. Objectives	
	1.1.3. Learning outcome.....	
	1.1.4. Photographs.....	
	1.1.5. Feedback of the day -1.....	
	Day – 2 report	
	1.2.1 TVS.....	
	1.2.2 Objectives	
	1.2.3 Learning outcome.....	
	1.2.4 Photographs.....	
	1.2.5 Feedback of the day -2.....	
	Day – 3 report	
	1.3.1 Hands on VIT.....	
	1.3.2 Objectives.....	
	1.3.3 Learning outcome.....	
	1.3.4 Photographs.....	
	1.3.5 Feedback of the day -3.....	

Day – 4 report

- 1.4.1 BSNL.....
- 1.4.2 Objectives
- 1.4.3 Learning outcome.....
- 1.4.4 Photographs.....
- 1.4.5 Feedback of the day -4.....

Day – 5 report

- 1.5.1 NIOT.....
- 1.5.2 Objectives
- 1.5.3 Learning outcome.....
- 1.5.4 Photographs.....
- 1.5.5 Feedback of the day -5.....

Bangalore

Day – 1 report

- 2.1.1 Cranes Varsity.....
- 2.1.2 Objectives
- 2.1.3 Learning outcome.....
- 2.1.4 Photographs.....
- 2.1.5 Feedback of the day -1.....

Day – 2 report

- 2.2.1 Cranes Varsity.....
- 2.2.2 Objectives
- 2.2.3 Learning outcome.....
- 2.2.4 Photographs.....
- 2.2.5 Feedback of the day -2.....

Day – 3 report

- 2.3.1 NeoSky.....
- 2.3.2 Objectives
- 2.3.3 Learning outcome.....

- 2.3.4 Photographs.....
- 2.3.5 Feedback of the day -3.....

Day – 4 report

- 2.4.1 Visvesvaraya Industrial Museum.....
- 2.4.2 Objectives
- 2.4.3 Learning outcome.....
- 2.4.4 Photographs.....
- 2.4.5 Feedback of the day -4.....

Day – 5 report

- 2.5.1 Tech Talk.....
- 2.5.2 Objectives
- 2.5.3 Learning outcome.....
- 2.5.4 Photographs.....
- 2.5.5 Feedback of the day -5.....

Conclusions

Day – 1 Report

1.1.1 Industry Name

VIT Chennai – Expert Talk by Mr. Madhan Kumar Srinivasan

1.1.2 Objectives

To learn from the real-life experiences and career journey of an industry professional.

To understand the challenges, efforts, and milestones involved in building a successful

IT career.

To get inspired by personal achievements, struggles, and success stories in the corporate

world.

To relate academic learning with real-world professional growth.

To understand how determination, consistency, and skill-building impact long-term career development.

1.1.3 Learning Outcome

During the session, **Mr. Madhan Kumar Srinivasan** shared his complete life journey from his **childhood and education** to **entering the IT industry and progressing to Infosys**. Major learnings include:

Understanding the role of humble beginnings and how dedication can transform one's

life regardless of background.

Importance of continuous learning, upskilling, and adapting to new technologies in

the IT field.

Realization that career growth is not instant, but built through years of consistent effort, discipline, and self-belief.

Learning from failures, personal struggles, and how to convert them into stepping stones

for success.

Understanding work culture at Infosys through his real experiences and responsibilities.

The session showed how **soft skills, communication, attitude, and perseverance** play a major role in achieving long-term career goals.

Gained motivation to plan my own career path with clarity and focus.

1.1.4 Photographs





1.1.5 Feedback

The session was extremely **motivational and inspiring**. Listening to Mr. Madhan Kumar Srinivasan's life journey—from his birthplace, childhood challenges,

educational struggles, and early career experiences to finally achieving a stable and successful role at Infosys—was impactful. His achievements and personal stories taught valuable lessons about hard work, patience, and continuous self-improvement. Overall, Day-1 set a positive tone for the entire experiential learning program.

Day – 2 Report

1.2.1 Industry Name

TVS – Ambattur Industrial Estate, Chennai

1.2.2 Objectives

- To understand the complete manufacturing and assembly process of two-wheelers at TVS.
- To observe automation, robotics, and lean manufacturing techniques used in the automotive industry.
- To gain knowledge of quality inspection systems and safety standards implemented in production units.
- To understand the supply chain workflow—from raw material handling to finished vehicle dispatch.
- To interact with industry professionals and learn about modern trends in automobile engineering.

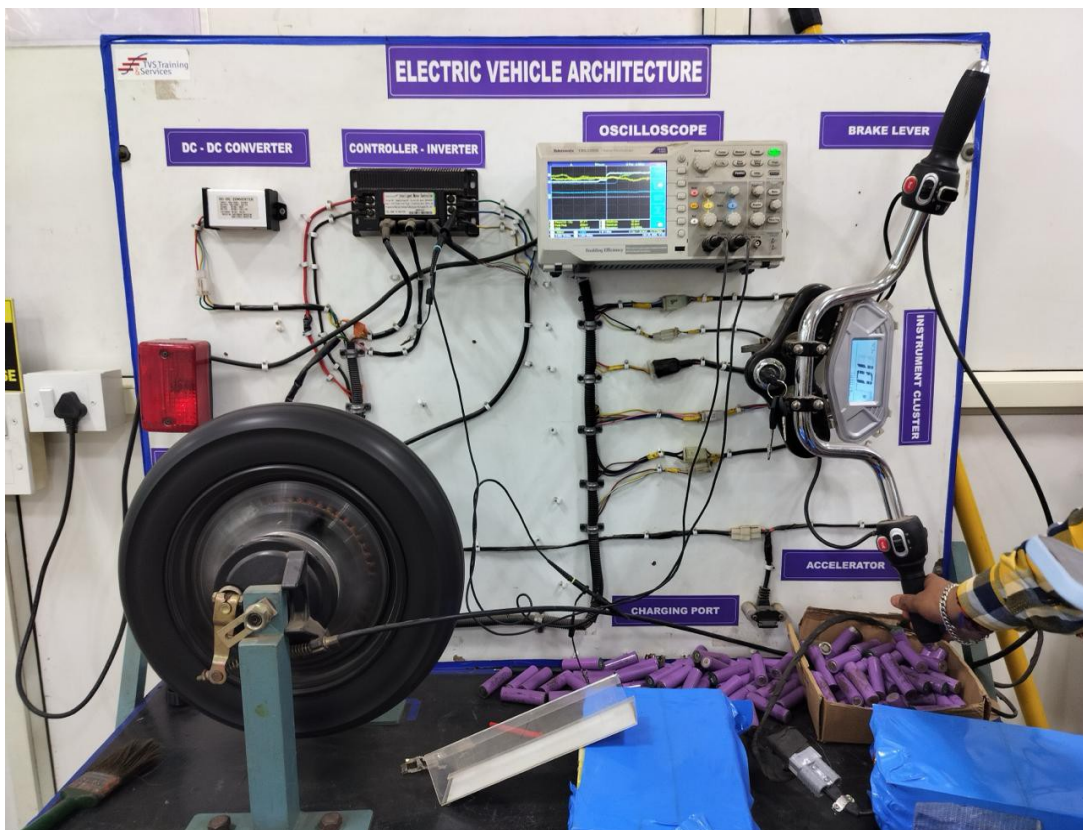
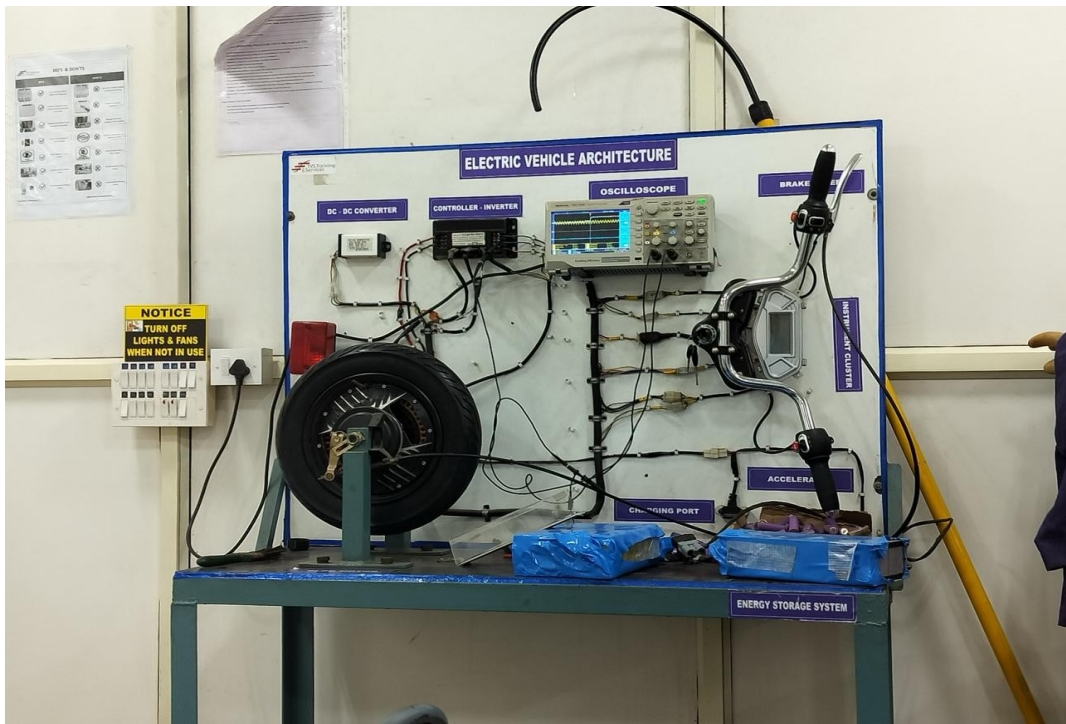
1.2.3 Learning Outcome

The industrial visit to the TVS manufacturing unit provided real-time exposure to large-scale automotive production. Major learning outcomes include:

- Understood the **full assembly line process**, including chassis preparation, engine assembly, painting, wiring, brake systems, and final testing.
- Observed the use of **automation and robotic systems** that enhance precision, efficiency, and safety in manufacturing.
- Learned about **lean manufacturing principles**, which reduce waste, improve workflow, and maintain production quality.
- Gained insights into **quality control mechanisms**, such as endurance testing, emission checks, vibration analysis, and safety validations.
- Observed how the factory maintains **inventory control**, spare parts management, and logistics coordination.
- Understood the importance of workplace safety measures, proper equipment handling, and strict process protocols.
- Developed a clear understanding of how modern automobile industries ensure performance, reliability, and customer satisfaction.

1.2.4 Photographs





1.2.5 Feedback

The visit to TVS was highly informative and engaging. Seeing the live production of two-wheelers and understanding the precision involved at each stage was an eye-opening experience. The use of automation, strict quality standards, and the well-structured assembly line demonstrated the advanced engineering and efficiency of TVS. Overall, Day-4 provided valuable practical exposure to the automotive industry and strengthened my interest in industrial engineering and manufacturing systems.

Day – 3 Report

1.3.1 Industry Name

VIT Chennai – Internet of Things (IoT) Technical Session

1.3.2 Objectives

- To understand the fundamentals and architecture of the Internet of Things.
- To gain exposure to IoT components such as sensors, microcontrollers, communication modules, and cloud integration.
- To explore real-time IoT applications used in industries and everyday life.
- To understand how IoT enables automation, data collection, and smart decision-making.
- To participate in demonstrations showcasing IoT-based systems and workflows.

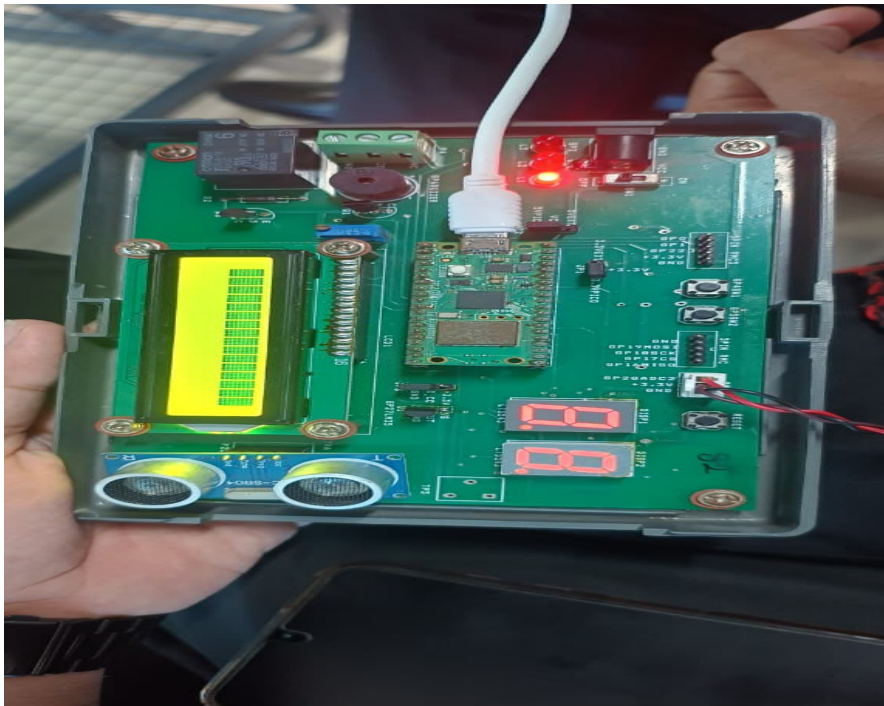
1.2.3 Learning Outcome

The IoT session provided practical and conceptual insights into how connected devices operate and communicate. Key learning outcomes include:

- **Understanding IoT Architecture** – including sensors, edge devices, gateways, and cloud platforms.
- **Exposure to Hardware** – such as Arduino, NodeMCU/ESP8266, Raspberry Pi, and various environmental sensors.
- Learned how sensors capture data and how microcontrollers process and transmit it.

- Understood different **communication protocols** such as Wi-Fi, Bluetooth, MQTT, and HTTP.
- Gained knowledge on how IoT is implemented in **smart homes, healthcare, agriculture, manufacturing, and transportation**.
- Understood challenges like security, interoperability, network reliability, and energy management.
- Developed interest in building small IoT prototypes for future projects and internships.
- Realized the importance of IoT in the upcoming era of automation and Industry 4.0.

1.3.4 Photographs





1.3.5 Feedback

The IoT session was highly interactive and informative. The demonstrations and real-time examples helped me clearly understand how connected devices work in real environments. The instructors explained concepts in a simple and engaging manner, making it easy to relate them to real-world applications. Overall, the session boosted my interest in IoT and motivated me to explore embedded systems and automation technologies further

Day – 4 Report

1.4.1 Event/Industry Name

RGMTTC BSNL (Regional Telecom Training Centre)

Location: Chennai

1.5.2 Objectives

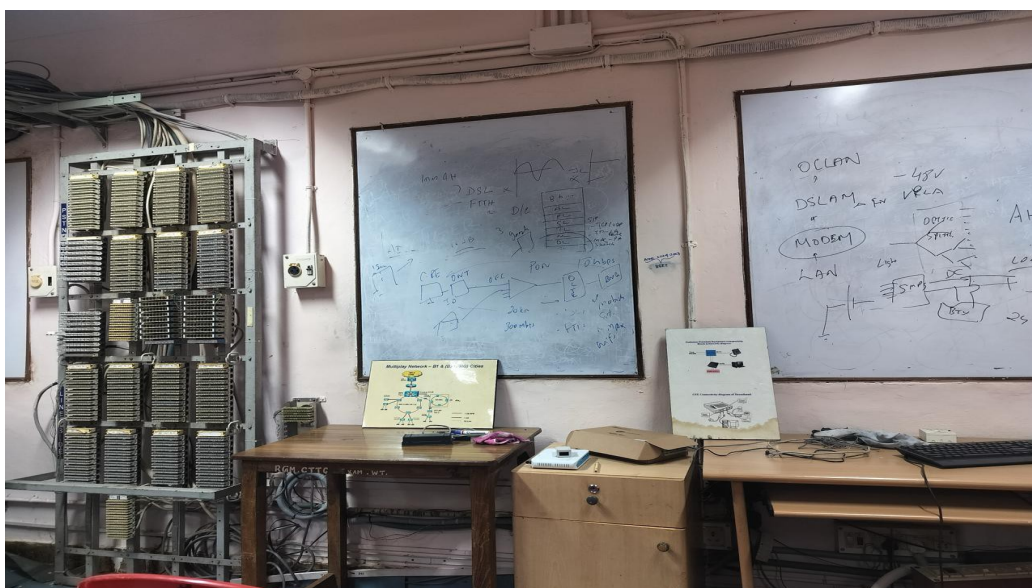
- To understand the fundamentals of telecommunication networks and digital communication systems.
- To learn about BSNL's infrastructure for mobile networks, broadband services, and optical fiber communication.
- To observe real-world switching, routing, and signal transmission operations.
- To gain awareness of telecom technologies used in India including GSM, LTE, FTTH, and VoIP systems.
- To interact with BSNL engineers and understand career opportunities in the telecom sector.

1.4.3 Learning Outcome

The visit to RGMTTC–BSNL provided extensive knowledge of how telecommunication networks operate at a national level. Key learning outcomes include:

- Understood the **architecture of telecom networks**, including base stations, switching centers, and backbone networks.
- Learned how **optical fiber communication (OFC)** works, including light propagation, attenuation, splicing, and testing techniques.
- Observed how **GSM and LTE networks** manage calls, SMS, and mobile data using SIM authentication, frequency allocation, and cell towers.
- Gained knowledge about **broadband technologies**, such as DSL, FTTH (Fiber-to-the-Home), and leased line services.
- Understood how BSNL manages **routing and switching**, including PSTN, IP networks, and signaling systems like SS7.
- Observed the operation of **exchange systems**, network monitoring tools, and real-time troubleshooting processes.
- Learned about security practices, redundancy planning, and backup mechanisms used to keep telecom services running 24/7.

1.4.4 Photographs







1.4.5 Feedback

The training session at RGMTTC–BSNL was extremely enriching and technically valuable. It provided a clear understanding of how communication networks function behind the scenes, from mobile connectivity to broadband operations. The engineers explained complex concepts in a simple and structured manner, making the session highly engaging. Overall, Day–5 significantly enhanced my understanding of modern telecom systems and marked a strong conclusion to the Chennai segment of the experiential learning program.

Day – 5 report

1.5.1 Industry Name

NIOT (National Institute of Ocean Technology)

1.5.2 Objectives

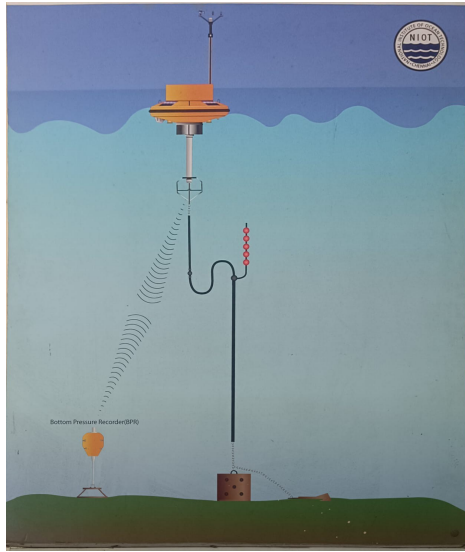
- To learn about India's indigenous marine technology and ocean research initiatives. .
- To understand engineering applications in oceanography, deep-sea exploration, and marine resource utilization
- To explore technologies related to desalination, deep-sea mining, ocean observation, and environmental monitoring.
- To interact with experts working on national-level marine projects and understand real-world challenges in ocean technology.
- To gain exposure to technological systems used for coastal safety, climate studies, and disaster management.

1.5.3 Learning Outcome

- Gained insights into key divisions such as **Ocean Electronics, Deep Sea Technologies, Coastal Engineering**, and **Energy & Freshwater** groups
- Understood the working of **Low-Temperature Thermal Desalination (LTTD)** technology used to produce fresh water from seawater.
- Learned about deep-sea exploration systems, remotely operated vehicles (ROVs), and autonomous underwater vehicles (AUVs).
- Understood how **tsunami early warning systems, data buoys, and ocean observation networks** help in disaster prediction and coastal safety.
- Gained exposure to offshore structures, environmental studies, and marine energy solutions like wave and current energy systems.
- Developed awareness of India's advancements in ocean technology and its importance for national development.

1.5.4 Photographs





1.5.5 Feedback of the Day – 3

The NIOT visit was extremely informative and eye-opening. Understanding India's progress in marine research and deep-sea technology gave a new perspective on the importance of ocean engineering. The advanced desalination systems, disaster warning technologies, and deep-sea exploration tools were fascinating. Overall, it was a highly educational and inspiring session that showcased how engineering can drive ocean-based innovation.

CHOICE-2

BANGALORE

Day – 1 Report

1.1.1 Industry Name

Cranes Varsity (Software Training Institute), Bangalore

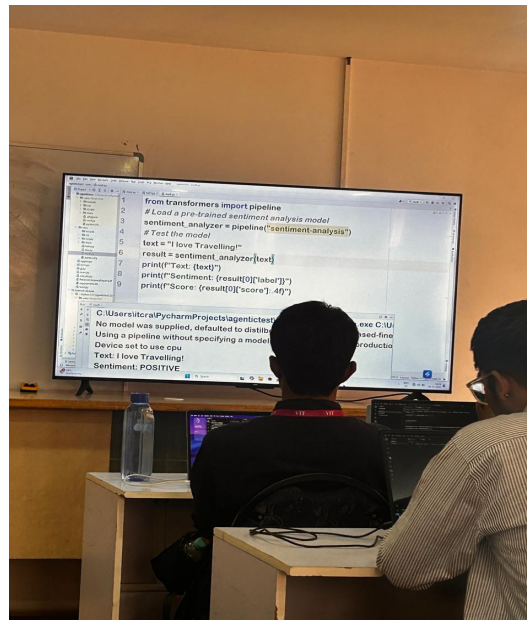
1.1.2 Objectives

- To gain exposure to industry-oriented training in Machine Learning, Deep Learning, and Large Language Models. .
- To understand the end-to-end workflow of building, training, and evaluating ML/DL models .
- To develop hands-on experience by training our own machine learning model under expert guidance.
- To understand how AI technologies are applied in modern software industries.
- To interact with trainers and learn about real-world tools, frameworks, and industry expectations.

1.1.3 Learning Outcome

- Learned the fundamentals of Machine Learning including supervised and unsupervised learning concepts .
- Understood essential algorithms such as linear regression, classification models, clustering, and evaluation metrics .
- Gained exposure to Deep Learning basics, neural networks, activation functions, and backpropagation.
- Learned about industry-relevant tools such as Python, NumPy, Pandas, Scikit-Learn, TensorFlow/PyTorch.
- Understood dataset preparation, preprocessing techniques, and split methods (train/test/validation).
- Became familiar with the workflow required to train and validate our own ML model.
- Gained insights into how industries use ML models for automation, prediction, and decision-making.
- Understood career paths in AI/ML and the skills required for professional opportunities.

1.1.4 Photographs



1.1.5 Feedback

The first day at Cranes Varsity was highly engaging and informative. The trainers explained Machine Learning concepts in a very clear and industry-oriented manner. The

practical demonstrations helped deepen our understanding of how ML algorithms work behind the scenes. Getting hands-on experience boosted our confidence and made the session truly interactive. Overall, Day 1 was an excellent introduction to real-world AI and ML applications.

Day – 2 Report

1.2.1 Industry Name

Cranes Varsity, Bangalore – ML, Deep Learning & LLM Hands-On Session

1.2.2 Objectives

- To gain a deeper understanding of Deep Learning architectures and real-world applications.
- To explore neural networks, CNNs, RNNs, and the fundamentals of modern AI systems
- To understand how Large Language Models (LLMs) work internally—including embeddings, transformers, and attention mechanisms.
- To perform hands-on training of an AI/ML model and evaluate its accuracy.
- To understand how models are deployed, optimized, and used in practical scenarios.
- To complete an assessment/test to earn the Cranest Varsity certification.

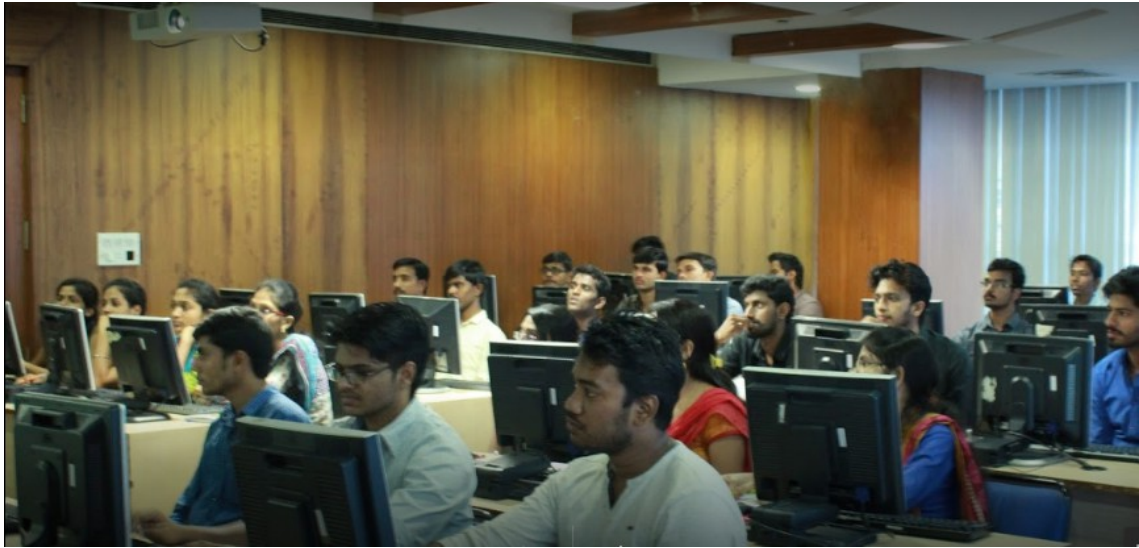
1.2.3 Learning Outcome

The second day focused heavily on advanced AI concepts and practical exposure. Key learnings include:

- Understood Deep Learning fundamentals, neural network layers, optimizers, and loss functions. .
- Learned how CNNs are applied in image processing and RNNs/LSTMs in sequence modelling .
- Gained introductory understanding of Large Language Models and modern transformer architecture .
- Explored concepts like attention mechanism, tokenization, embeddings, and inference.
- Understood how LLMs power applications such as chatbots, summarizers, text generation, and search systems.
- Successfully trained our own ML model and validated results through accuracy metrics.
- Learned about model improvement techniques, hyperparameter tuning, and dataset augmentation.
- Developed confidence in applying ML and DL concepts for real projects.

1.2.4 Photographs

(*taken from the internet)



1.2.5 Feedback of the Day – 2

The second day was extremely enriching and technical. Understanding Deep Learning and LLM architectures helped bridge the gap between theory and industry practices. The hands-on model training was one of the most exciting parts of the entire visit. The instructors explained complex topics like transformers and attention mechanisms in an easy-to-understand way. The certification test further strengthened our learning, making Day 2 highly productive and inspiring.

Day – 3 Report

1.3.1 Industry Name

**Neosky – Drone Remote Pilot Training Organisation,
Bangalore**

1.3.2 Objectives

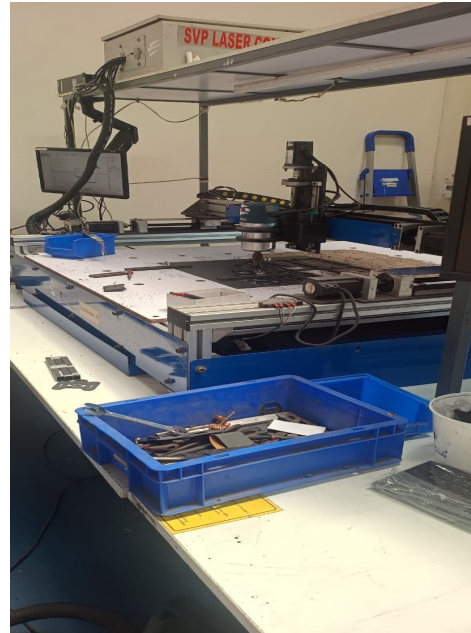
- To understand the role and applications of drones in modern industries .
- To learn how drones are designed, built, and controlled.
- To explore the physics, aerodynamics, and flight mechanisms behind UAVs.
- To observe practical drone demonstrations in open-ground conditions.
- To understand how drone technology benefits sectors like agriculture, medical services, logistics, defence, and media.

1.3.3 Learning Outcome

- Gained knowledge of drone components, materials used, and the engineering behind UAV construction.
- Understood flight principles such as thrust, lift, drag, and stability control.
- Learned about various real-world applications of drones—medical delivery, crop monitoring, surveillance, military reconnaissance, mapping, and cinematography.
- Understood safety regulations, operational guidelines, and the importance of certified drone pilots.
- Experienced a live drone flying demonstration in open ground, observing take-off, maneuvering, and landing techniques.
- Developed awareness of career opportunities in drone piloting, UAV maintenance, and aerial data analytics.

1.3.4 Photographs





1.3.5 Feedback of the Day – 3

The Neosky visit was exciting and highly informative. Learning the science and engineering behind drones helped us understand how impactful UAVs are across different fields. Watching the live drone flying session made the experience even more engaging. The session sparked curiosity about drone piloting, engineering, and innovative aerial solutions .

Day – 4 Report

1.4.1 Industry Name

Bangalore Museum / Science & Technology Museum

1.4.2 Objectives

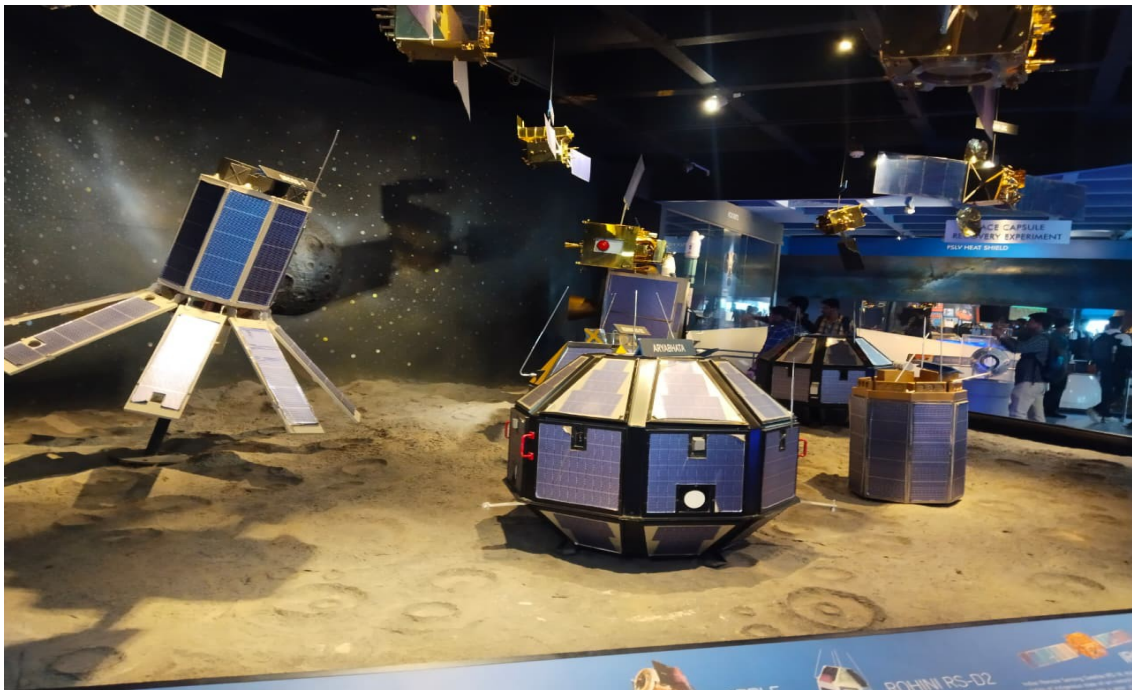
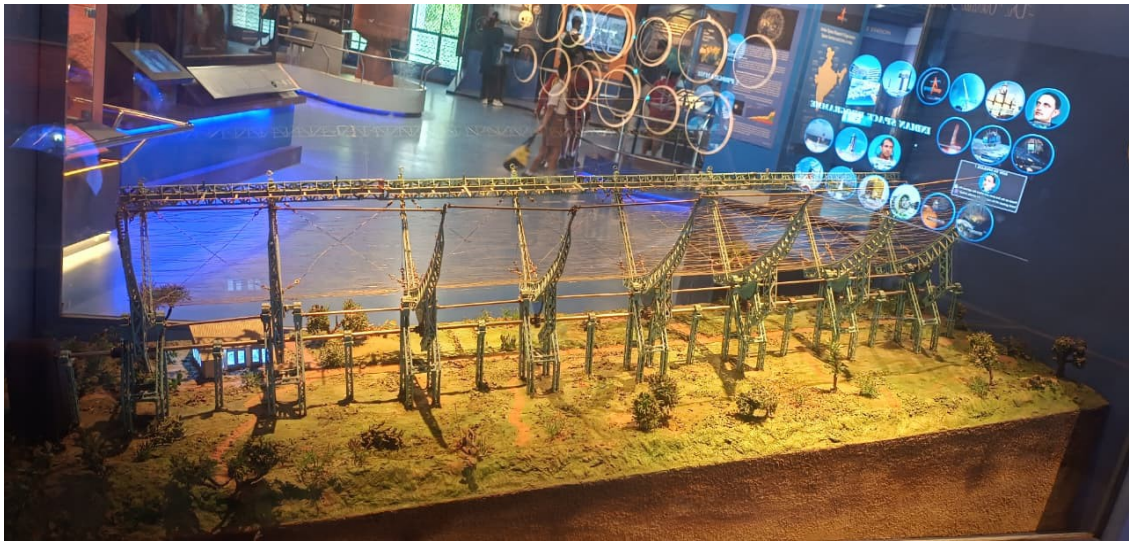
- To learn about historical advancements in science and engineering.
- To observe scientific models demonstrating mechanical, electrical, optical, and physical principles.
- To understand how technology evolved from early inventions to the modern era.

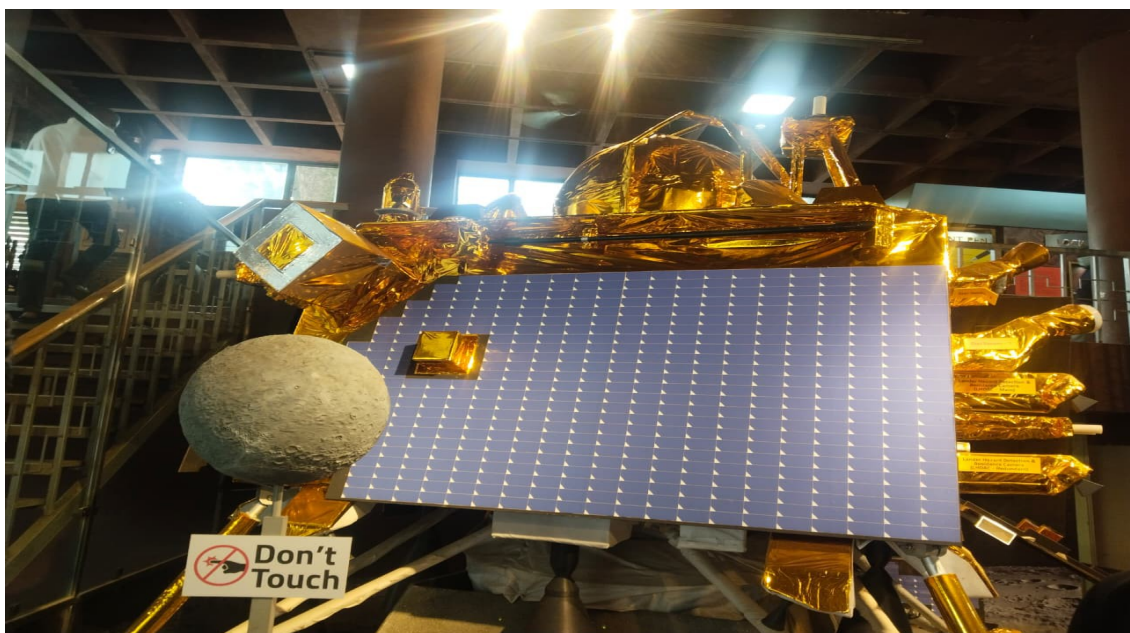
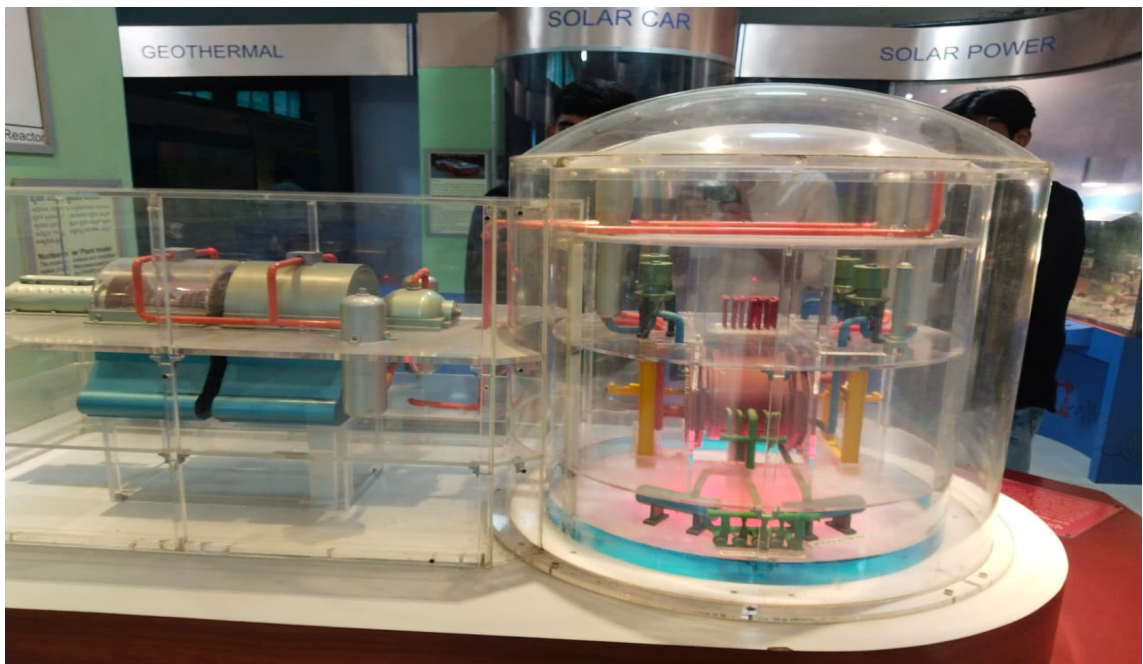
1.4.3 Learning Outcome

- Learned about early scientific instruments, telescopes, engines, and communication devices.
- Understood practical working models of **electric motors, pulleys, hydraulics, optics, energy generation**, etc.
- Observed interactive displays explaining physics, maths, and engineering concepts.
- Learned how innovation developed over centuries and shaped modern technology.
- Gained motivation to explore science beyond textbooks using visual and hands-on learning.

1.4.4 Photographs







1.4.5 Feedback of the Day – 4

The museum visit was a very educational experience. The interactive exhibits and science models provided clear insights into engineering fundamentals. It was a fun and informative day that improved conceptual understanding.

Day – 5 Report

1.5.1 Event/Industry Name

Industry Expert Session – Mr. Krishnav Dave (Founder, PreProd Corp)

Topic: *Large Language Models (LLMs), AI Models & Generative AI*

1.5.2 Objectives

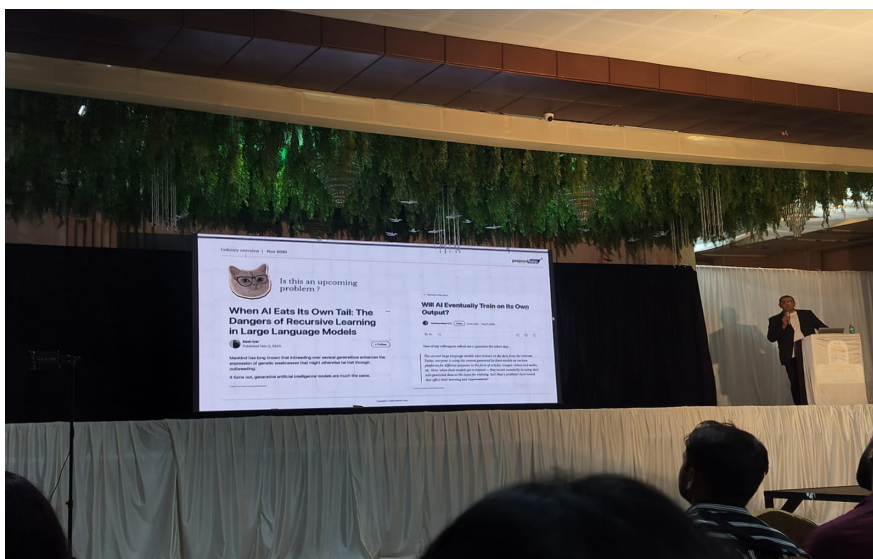
- To understand the role and evolution of **Large Language Models (LLMs)** in modern AI systems.
- To learn how generative AI is transforming industries such as software development, healthcare, finance, and cybersecurity.
- To gain insights into the architecture, training, and deployment of advanced AI models.
- To explore real-world case studies and industry applications built using LLMs.
- To receive guidance on skills required for building a career in Artificial Intelligence and Machine Learning.

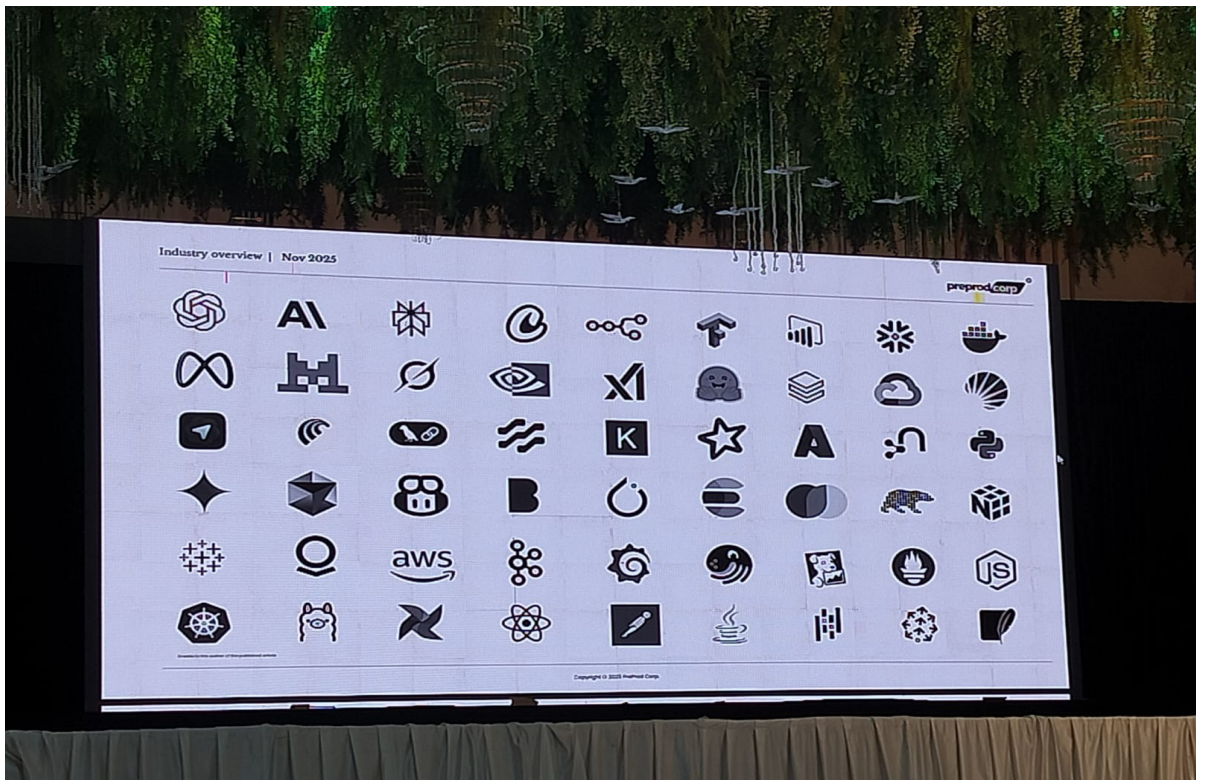
1.5.3 Learning Outcome

The session by **Mr. Krishnav Dave** provided deep technical and industry-level insights into the rapidly growing world of AI. Key learning outcomes include:

- Understood what **Large Language Models (LLMs)** are and how they are trained using billions of parameters and massive datasets.
- Learned about **transformer architecture**, attention mechanisms, and how models like GPT, LLaMA, and Gemini process and generate text.
- Gained clarity on various types of AI models:
 - **Language models**
 - **Vision models**
 - **Multimodal models**
 - **Generative models (GANs, Diffusion Models)**
- Understood real-world applications of GenAI such as:
 - Chatbots & virtual assistants
 - Code generation & debugging
 - Content creation (text, image, video)
 - Data analysis & summarization
 - Automation in enterprises
- Learned about challenges in AI: hallucinations, bias, safety, data privacy, model alignment, and responsible AI.
- Understood the future scope of GenAI in startups, enterprises, and research.
- Gained guidance on building skills in Python, ML libraries, prompting, model fine-tuning, vector databases, and MLOps.

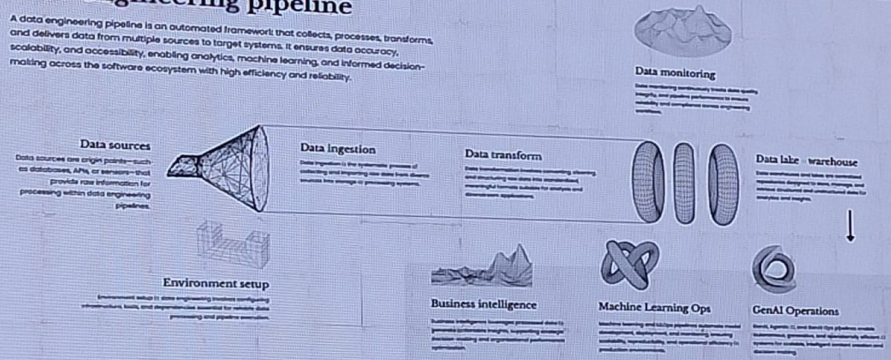
1.5.4 Photographs





Data engineering pipeline

A data engineering pipeline is an automated framework that collects, processes, transforms, and delivers data from multiple sources to target systems. It ensures data accuracy, scalability, and accessibility, enabling analytics, machine learning, and informed decision-making across the software ecosystem with high efficiency and reliability.



1.5.5 Feedback of the Day – 5

The expert session by **Mr. Krishnav Dave** was one of the most insightful experiences of the program. His explanations of LLMs and generative AI were easy to understand despite the complexity of the topic. The session gave me a clear understanding of how AI is shaping the future and how students can prepare for a career in AI and ML. It was a highly motivating and futuristic session, marking a strong conclusion to the Bangalore experiential learning journey.